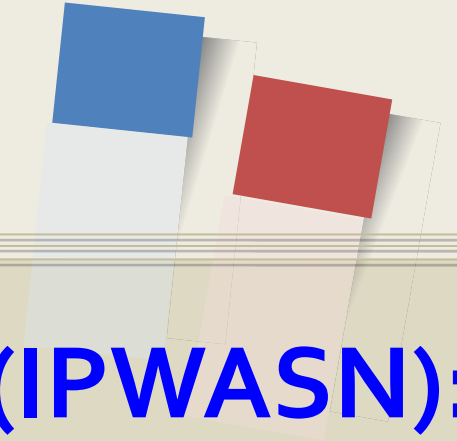


IETF-126 V6OPS Meeting



IPv6 Wireless Access in Satellite Networks (IPWASN): Problem Statement and Use Cases ([draft-lee-v6ops-ipv6-satellite-networks-00](#))

July 24, 2026

Vienna in Austria

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Introduction to IPWASN

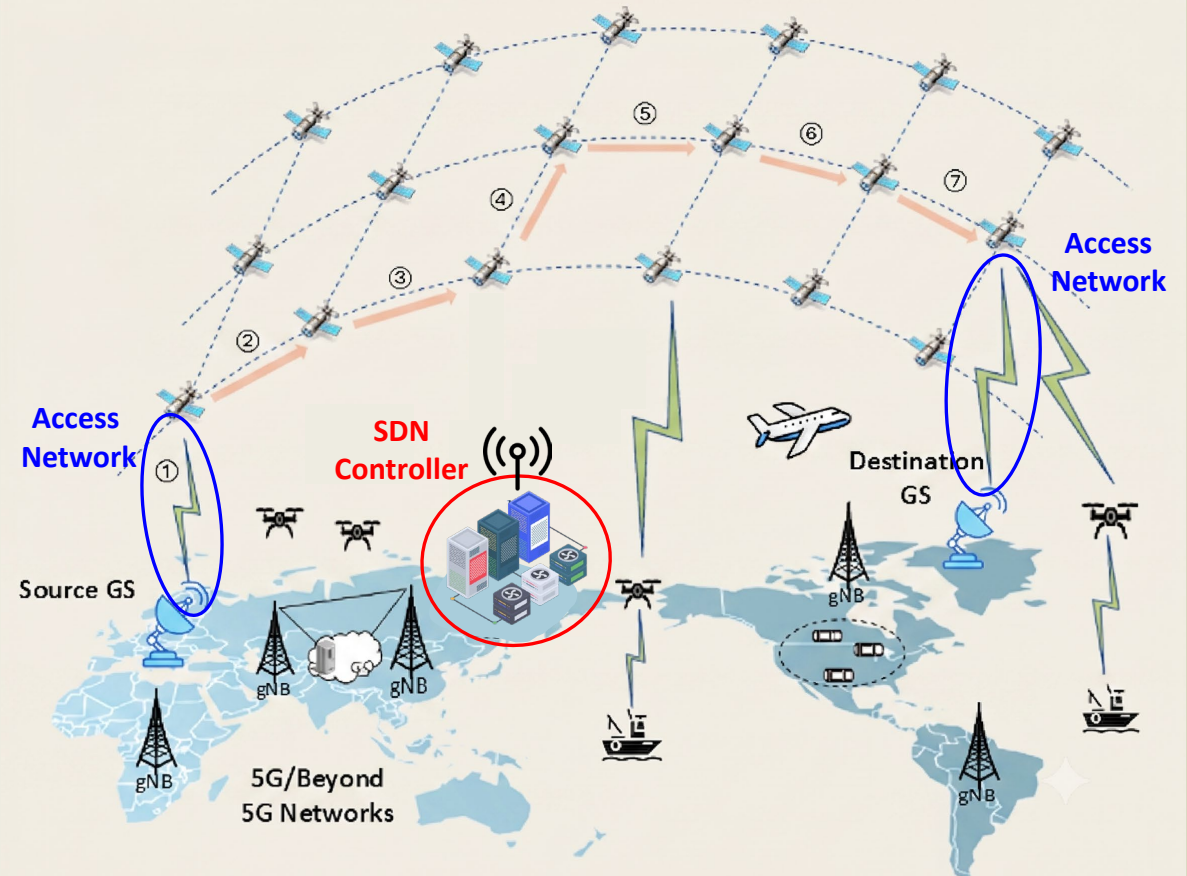


Overview

- To connect LEO satellites, ground stations, and terrestrial networks.
 - Uses Inter-Satellite Links (ISLs) for multi-hop transmission.

Objective

- To schedule paths based on satellites' movement and link availability.
- To improve handover performance between satellites for ISLs.
 - while maximizing packet delivery ratio and minimizing End-to-End (E2E) Delay.





Goal of IPWASN Draft

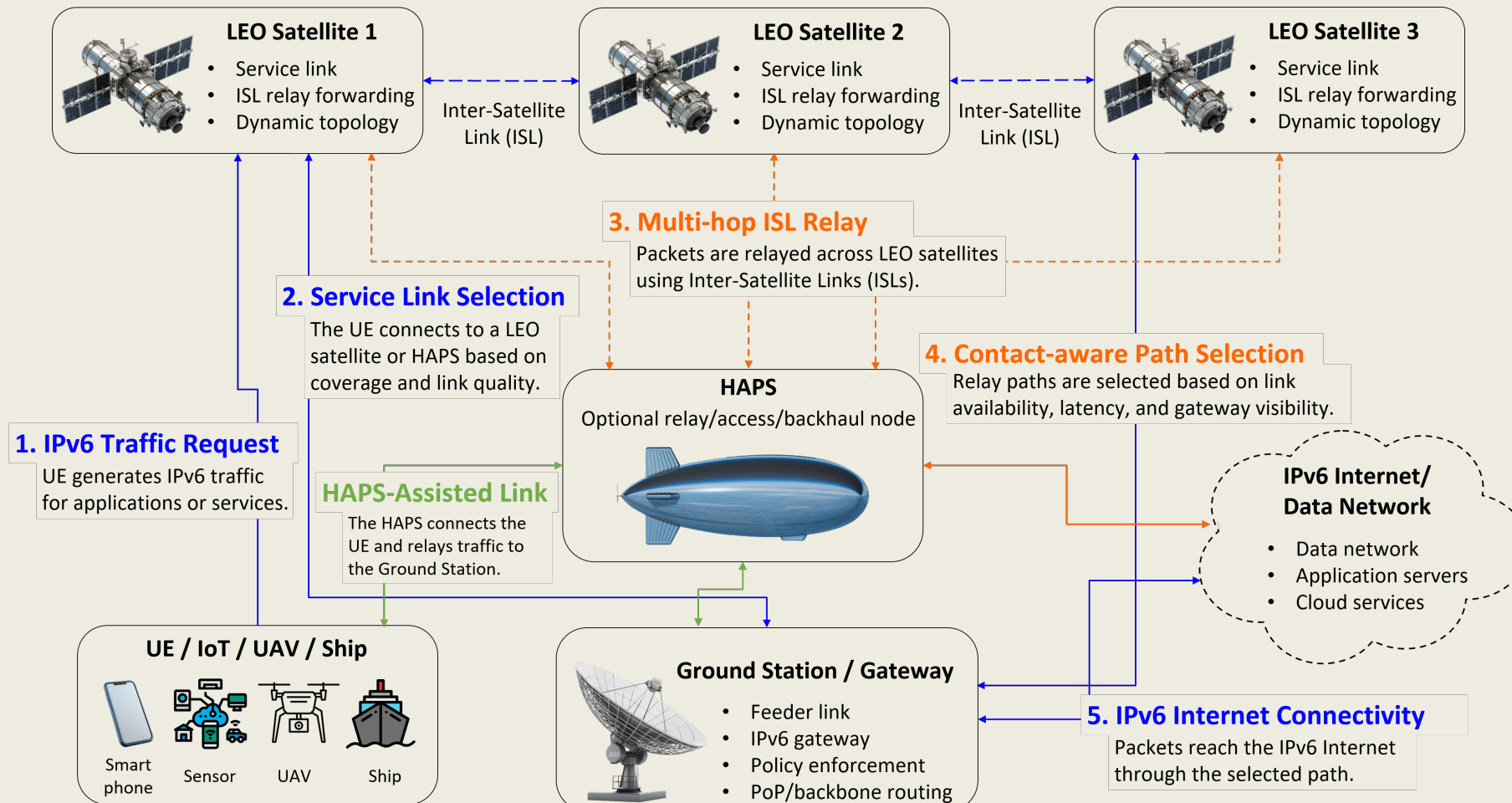
❑ Main Goal:

- ❑ **How to select a gNB (Base Station)?**
 - ❑ To select the best access point for IPv6-based NTN communication.
- ❑ **How to construct an Inter-Satellite Link (ISL) path from source Ground-Control Center (GCC) to destination GCC?**
 - ❑ To build a multi-hop satellite path between ground gateways.
- ❑ **IPWASN supports IPv6-based wireless access in Non-Terrestrial Networks (NTNs).**
 - ❑ Target networks include LEO satellite networks, Inter-Satellite Links (ISLs), HAPS-assisted relay paths, and ground gateways.



IPWASN Network Architecture

IPv6 Satellite Networks for Nomadic User Devices





Problem Statement



❑ Dynamic Topology and Route Instability

- ❑ LEO satellites and ISLs move or change continuously.
 - ❑ IPv6 routes may become invalid, requiring rapid route updates and alternative path selection.

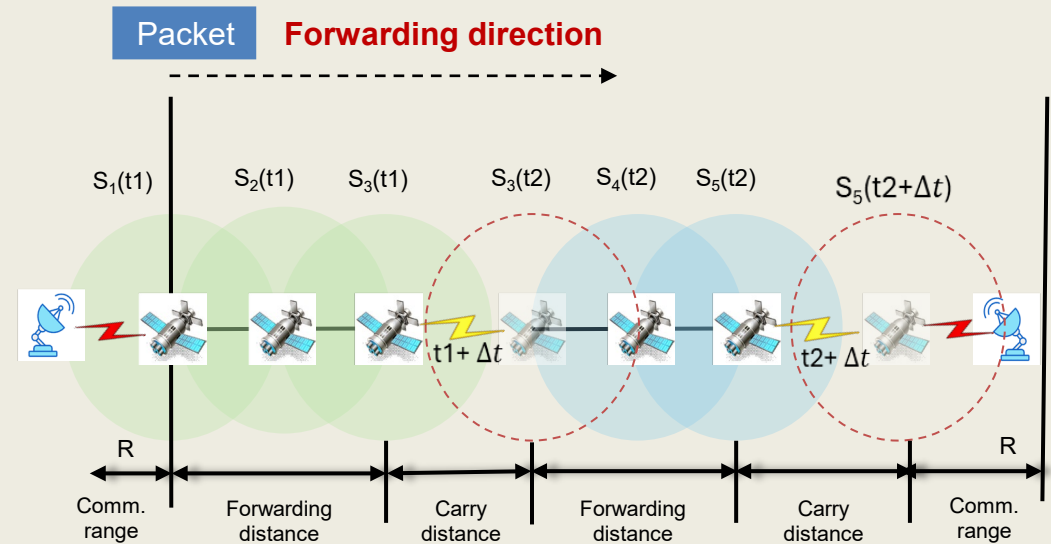
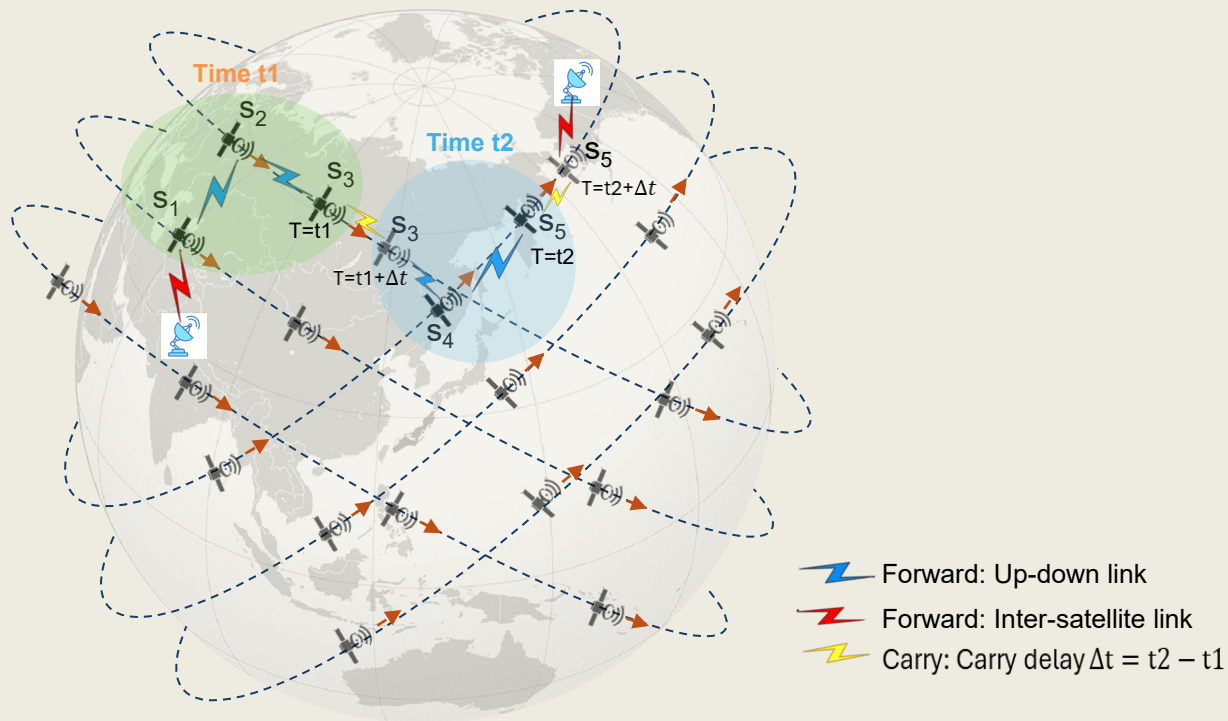
❑ Multi-Hop Forwarding and Intermittent Connectivity

- ❑ Packets may traverse multiple satellites and HAPS nodes before reaching a gateway.
 - ❑ Temporary link disruption may require an alternative relay, gateway, or store-and forward path.

❑ Mobility, QoS, and Traffic Engineering

- ❑ Users and satellite access points may change during an active communication session.
 - ❑ IPv6 mobility, QoS, and forwarding policies must adapt to handovers and time varying network conditions.

Packet Delivery through Packet Carry-and-Forward of Satellites



- ❑ Satellite mobility can change the forwarding path while user traffic is still active.
- ❑ IPv6 network connectivity needs to be preserved across satellite, gateway, and terrestrial network transitions.
- ❑ We need to design Access Networks and Core Networks of moving satellites.

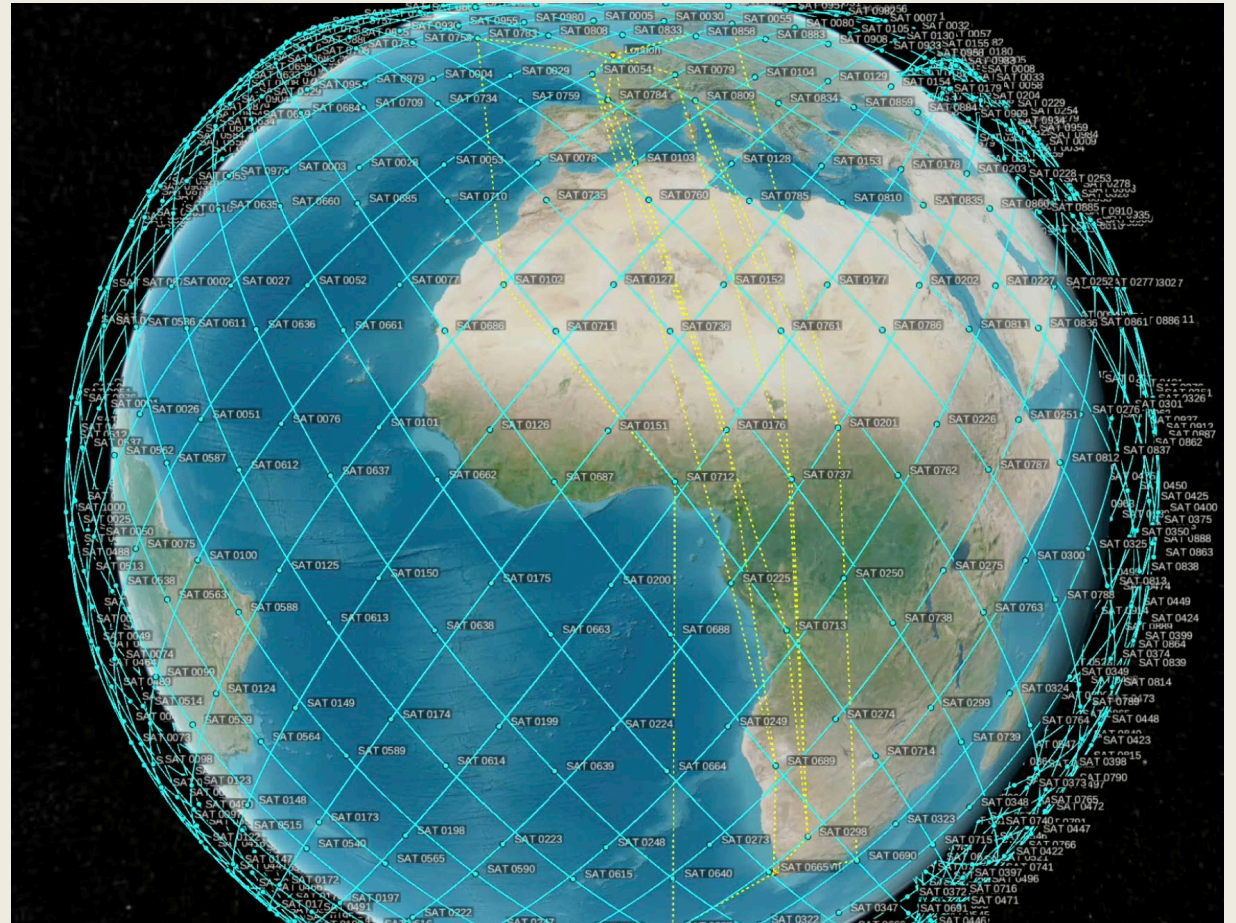
Operational Considerations for IPv6 Satellite Networks

Dynamic Topology of Core Networks

- Satellites' positions keep changing for forwarding paths.
- IPv6 routing should preserve service continuity considering this dynamic topology of satellites.

Satellite Link Conditions

- Long delay and intermittent links affect IPv6 operations.
- Key impacts include handover, routing updates, QoS, and transport behavior.





Next Steps



❑ To Refine the Problem Statement and Use Cases with Gap Analysis

- ❑ To clarify the reference scenarios, use cases, key requirements, and gap analysis.
 - ❑ Focus on dynamic topology, multi-hop forwarding, mobility, intermittent connectivity, and QoS.

❑ To Analyze and Evaluate Existing IPv6 Technologies

- ❑ To analyze the applicability of IPv6 mobility, routing protocols (including MANET), SRv6, QoS, and traffic-engineering mechanisms (e.g., MPLS).
 - ❑ Identifying technical gaps and interoperability requirements for routing continuity in underlay satellite networks with highly dynamic topology changes.

❑ To Provision Reliable Optimal End-to-End Paths

- ❑ To explore machine learning and quantum computing for provisioning reliable and optimal E2E paths between nomadic user devices over satellite core networks.
 - ❑ Considering multi-user traffic, multiple flows, satellite mobility, changing link states, and network/energy load balancing as optimization inputs.



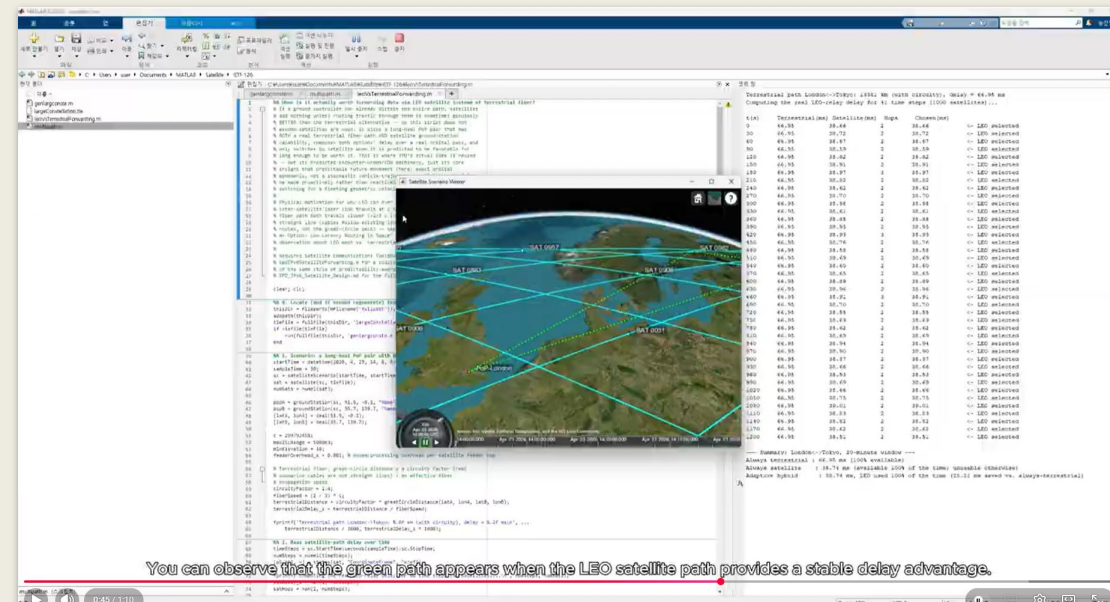
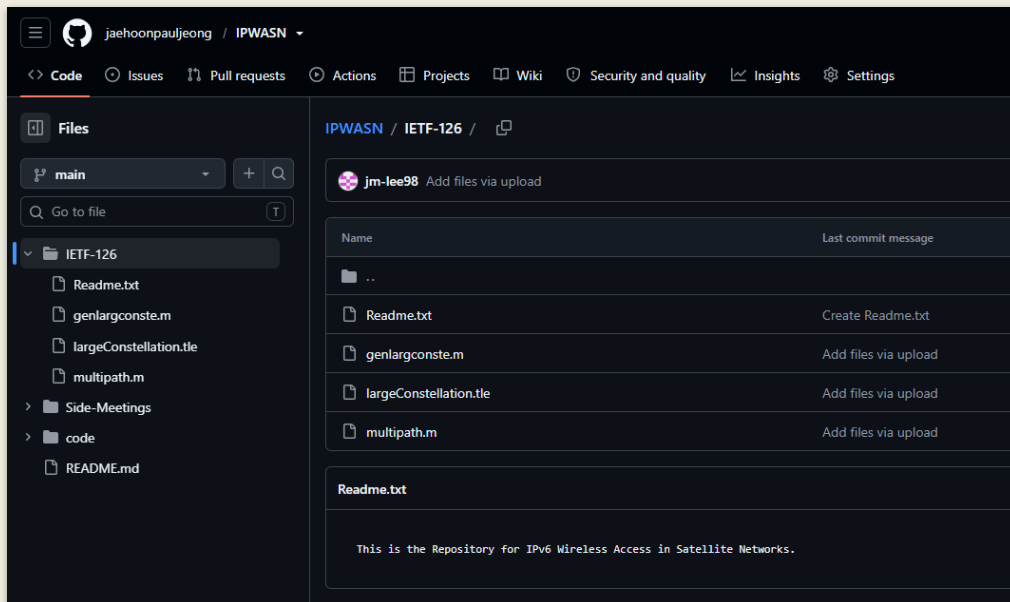
Appendix

❑ Open-Source Project for IPWASN

- ❑ [URL] <https://github.com/jaehoonpauljeong/IPWASN/tree/main/IETF-126>

❑ Demonstration Video Clip for IPWASN

- ❑ [URL] https://www.youtube.com/watch?v=3FXah_x0Ri4



Thank you and Any Comments?



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